

Multimedia Networking and Quality of Service



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Multimedia and QoS

- Audio/video Communication
- Virtual reality
- Streaming = continuous delivery of audio/video chunks over time
 - Different from browsing, messaging, file download...
- Real-time vs non-real-time streaming
 - live video and interactive audio are delay-sensitive
 - VoD is more throughput-sensitive
- QoS (Quality of Service) = how well the network meets app requirements, such as **delay, jitter, loss, throughput**
- Best-effort Internet: IP by default does not guarantee bandwidth, delay or loss bounds
- Key idea: QoE (user experience) matters more than raw QoS metrics, apps can trade quality vs stalls



Engineering QoS for multimedia

- **Overprovisioning** in the core and CDNs (Content Distribution Networks) close to users: shorter paths and fewer bottlenecks
- **Application-layer** techniques: HTTP adaptive streaming (DASH, HLS) with bitrate adaptation, buffering and prefetching
- What about network-layer mechanisms?



Streaming Apps Requirements

- Very low **one-way delay** (<150 ms), to keep conversation natural and controls responsive.
- **Delay variation** (jitter) must be small and smoothed by short playout buffers, otherwise audio and video become choppy or desynchronized.
- **Packet loss** must be limited, ideally below 1–2% for real-time media, because too many missing packets cause artifacts, frozen frames and audio glitches.
- Available **throughput** must exceed the media bitrate with some margin, so the application can adapt quality without frequent stalls.
- For non-interactive streaming (VoD, live events with a few seconds delay), larger buffers and slightly higher delay are acceptable, but loss and throughput still directly affect stall frequency and resolution.



Countermeasures in the Network

To support QoS:

- Traffic classification
- Packet scheduling algorithms
 - WFQ, RR, WRR, CBQ
- Control on traffic entering the network
 - At various levels

In other words:

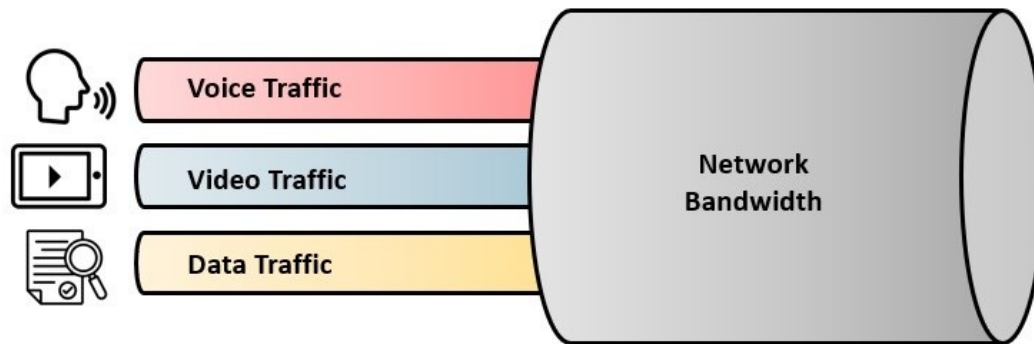
- Limit the number of packets arriving at network nodes
- Handle appropriately packets that need specific QoS



Traffic Classification

- Based on implementation contained in the IP and TCP/UDP headers (5-tuple):

- Destination IP address
- Source IP address
- Transport protocol
- Destination port
- Source port



- “Complex” algorithms
- Hardware implementations:
 - ASIC: Application Specific Integrated Circuit
 - CAM: Content Addressable Memory

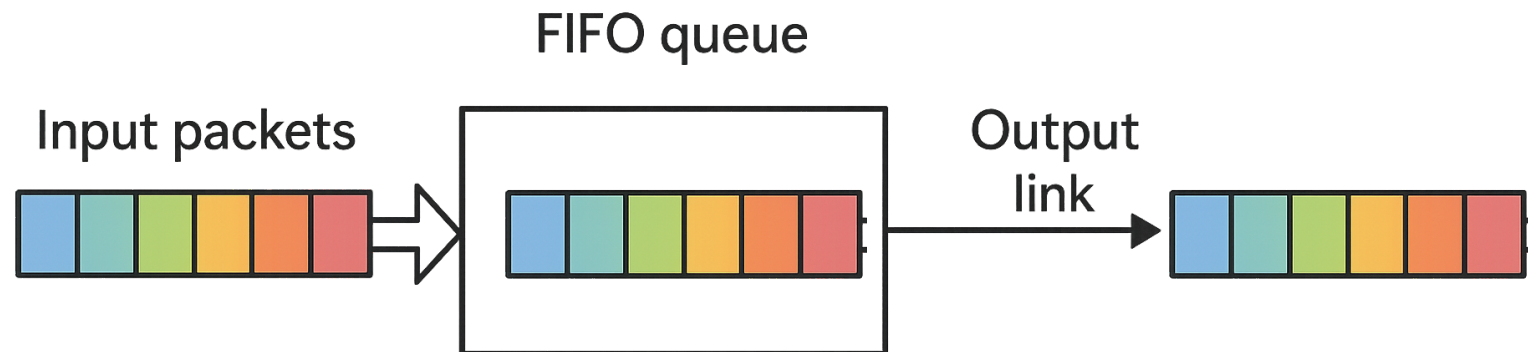


Packet Scheduling Algorithms

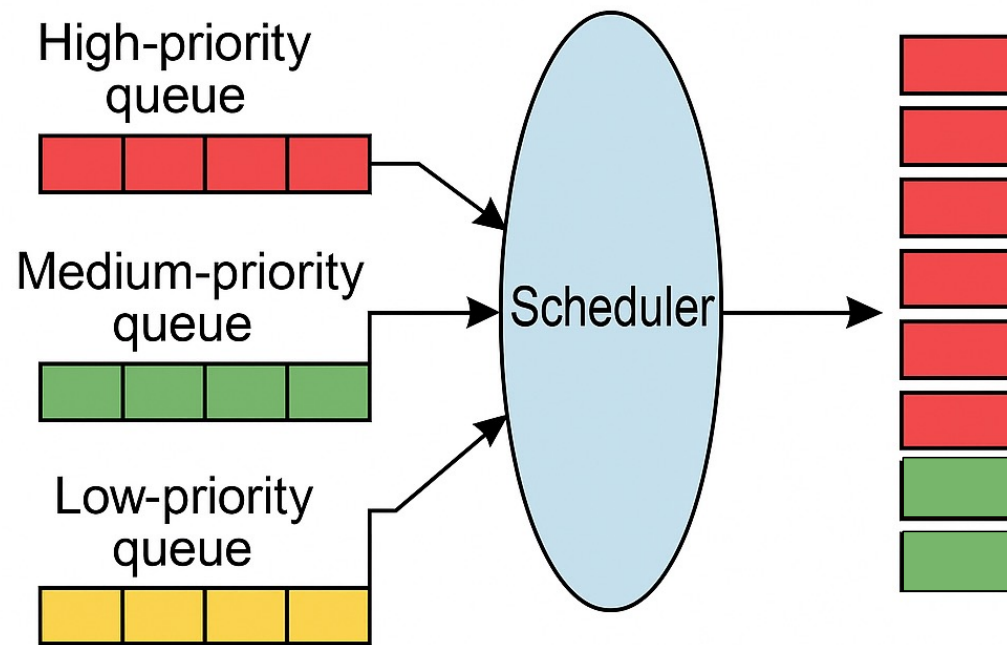
- Decides which packet is sent next on a link (who goes first)
- Executed on (some) routers
 - FIFO: simplest, no QoS control, all packets in one queue
 - Priority Queuing: higher-priority traffic served first, risk of starving low priority
 - Round Robin / Weighted RR: each queue gets turns, weights control bandwidth share
 - WFQ / DRR: approximate fair sharing, per-flow or per-class rate and delay control for QoS



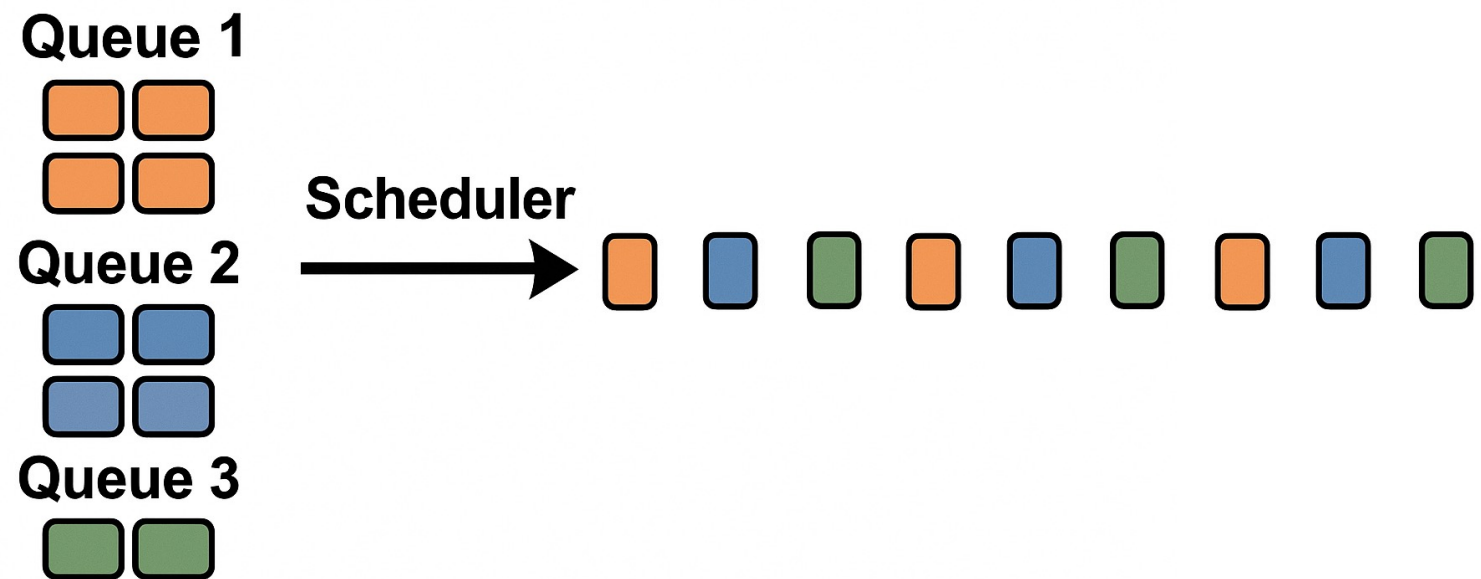
Simple Queuing: FIFO



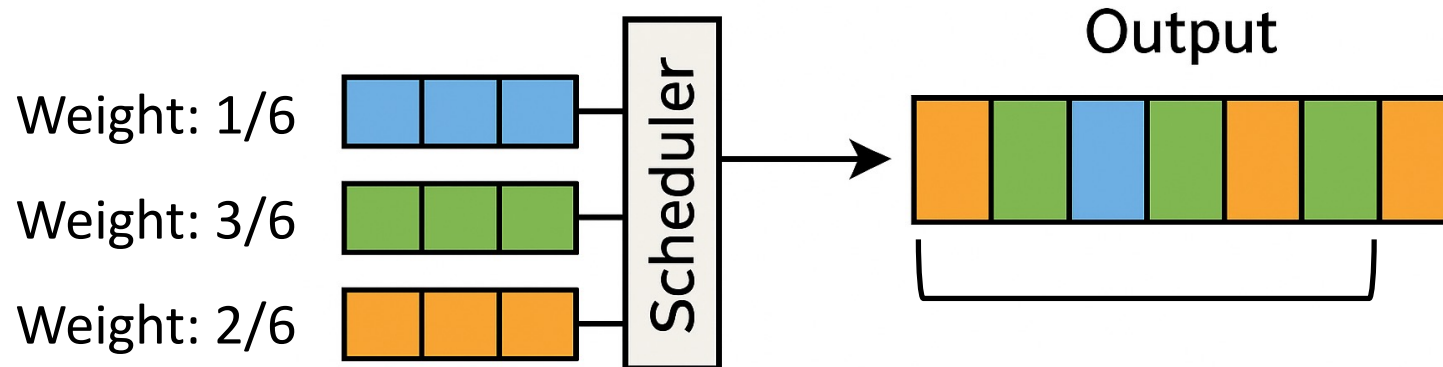
Priority Queueing



Round Robin

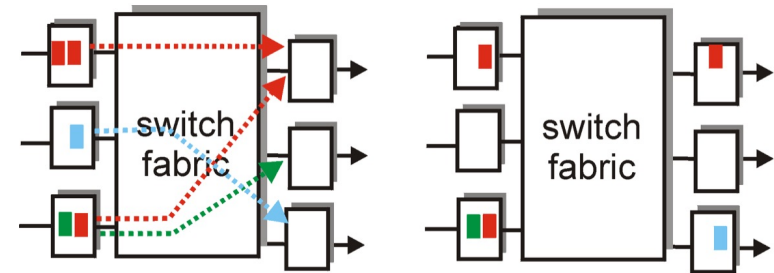


Weighted Fair Queueing



Note: switching capacity

- Guaranteed immediate switching requires speed-up in the switching fabric
 - the internal transfer rate of the fabric must be greater than the line rate of each input



- With speed-up, multiple inputs can simultaneously send to the same output, and the fabric can still serialize them within one external time slot.
- Without sufficient speed-up, contention at outputs causes internal queues, increased delay, and possible cell loss.
- The need for speed-up is especially critical at high-capacity links (e.g., 100 Gb/s, 400 Gb/s), where even small internal delays or inefficiencies quickly translate into large buffers and QoS degradation.

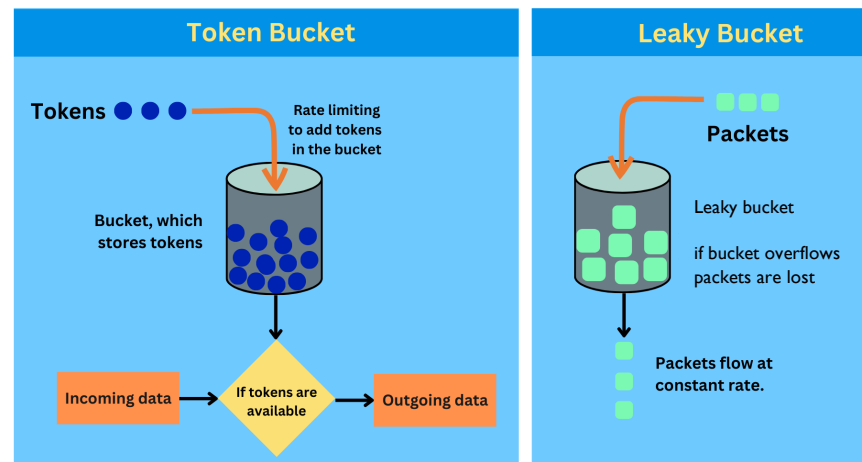
Control on Traffic entering the network

- Packet level
 - Shaping/policing
- A priori
 - Network engineering
 - Network dimensioning according to expected traffic
 - Limit the number of users
 - Traffic engineering
 - Controlled distribution of traffic across the network



Policing and shaping traffic

- Traffic policing and shaping control how flows use bandwidth, to enforce QoS and SLAs.
- **Token bucket**
 - allows bursts: tokens arrive in the bucket at rate R , up to capacity B ; a packet is sent or marked “in-profile” only if enough tokens are available, so average rate $\leq R$ but short bursts up to B are allowed.
- **Leaky bucket**
 - smooths traffic: packets (or credits) enter a bucket and are drained at a constant rate R ; if packets arrive faster than R , they are queued or dropped so the output becomes a nearly constant-rate stream.



Policing and shaping traffic

- Token bucket is typically used for policing at network edges (classify packets as in/out of contract, possibly drop or remark out-of-profile traffic).
- Leaky bucket is typically used for shaping, delaying packets in a buffer so that the flow respects a configured rate even when the source sends in bursts.
- In practice, many routers implement shapers as a token bucket followed by a leaky bucket, with tokens controlling dequeue rate and the queue absorbing bursts, combining the flexibility of token bucket with the smoothing effect of a leaky bucket.



Network Engineering/Traffic Engineering

Preventive actions:

- Network is dimensioned for the (almost) worst case
 - Statistics on user traffic
- Traffic matrix is determined
 - Traffic distribution

Actions throughout:

- Network state is continuously monitored
- Network dimensioning and traffic matrix can be changed if needed



Internet's Ambitious Solution: Integrated Services (IntServ)

- uses the Resource Reservation Protocol (RSVP) to signal QoS requirements along the end-to-end path, ensuring that each router can reserve the necessary resources
- maintains per-flow state information in routers, allowing for precise control and management of individual traffic flows
- admission control ensures that new flows are only admitted if sufficient resources are available, preventing resource contention and maintaining QoS guarantees
- offers multiple service classes, including guaranteed service for strict latency and bandwidth requirements,

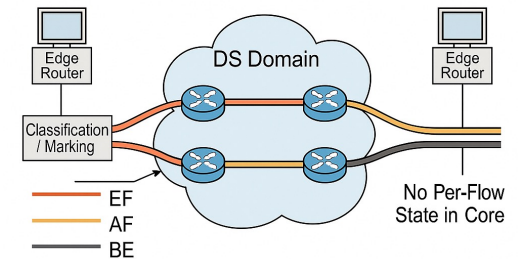
Too much overhead! – Not used



Lowering Ambitions: Differentiated Services (DiffServ)

- DiffServ uses an aggregate approach to QoS, classifying packets into broad categories rather than managing individual flows -> simpler network management
- When entering the DiffServ domain, packets are marked with a Differentiated Services Code Point (DSCP) using ToS in IPv4 and Type of Class in IPv6 headers
- Routers apply predefined per-hop behaviors (PHBs) based on these markings (via queueing/priority scheduling/WFQ)

Lighter and more manageable



Diffserv Today

- Inside ISP cores networks:
 - Packets are marked with a small set of DSCP values, mapped to a few behavior aggregates (e.g., EF for voice, AF classes for premium / assured traffic, BE for best effort)
- In enterprise and campus networks:
 - QoS policies for voice (VoIP), video, critical business apps.
 - Access switches mark/remark DSCP; WAN edge routers use them to prioritize during congestion.
- At the boundaries with other technologies:
 - Wi-Fi QoS (WMM) usually maps DSCP to access categories (different wait times before channel sensing)
 - Mobile networks (LTE/5G) often map IP DSCP to specific bearers, i.e., they use DSCP as input to internal QoS classes

